Python Virtual Environments



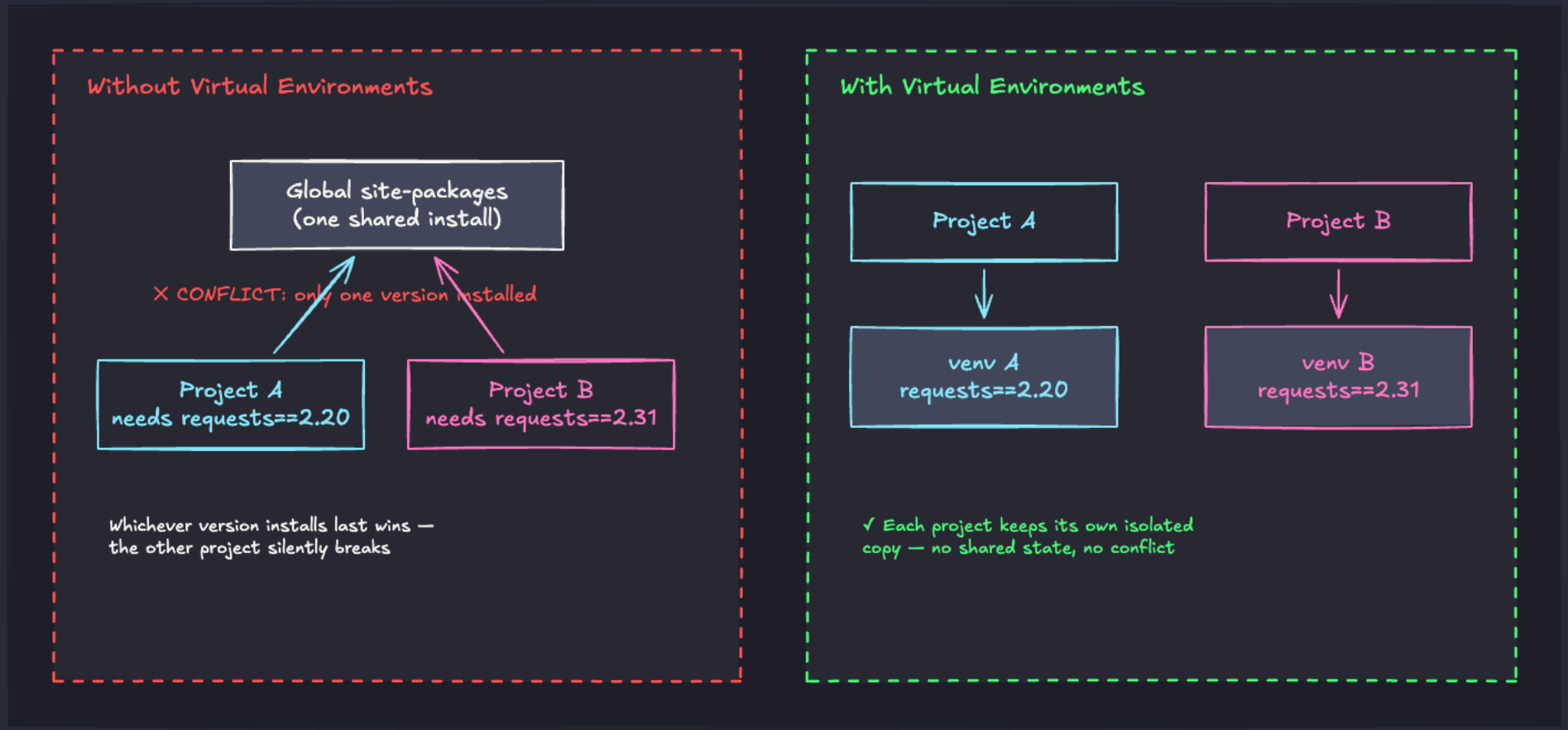
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The Problem: One Python, Many Projects

- Your machine has **one** system Python install by default
- Every project you `pip install` into shares that same package library
- Two projects rarely want the *same* version of the *same* package
- Symptom: code that worked last week now throws `ImportError` or version errors

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Without vs. With Virtual Environments



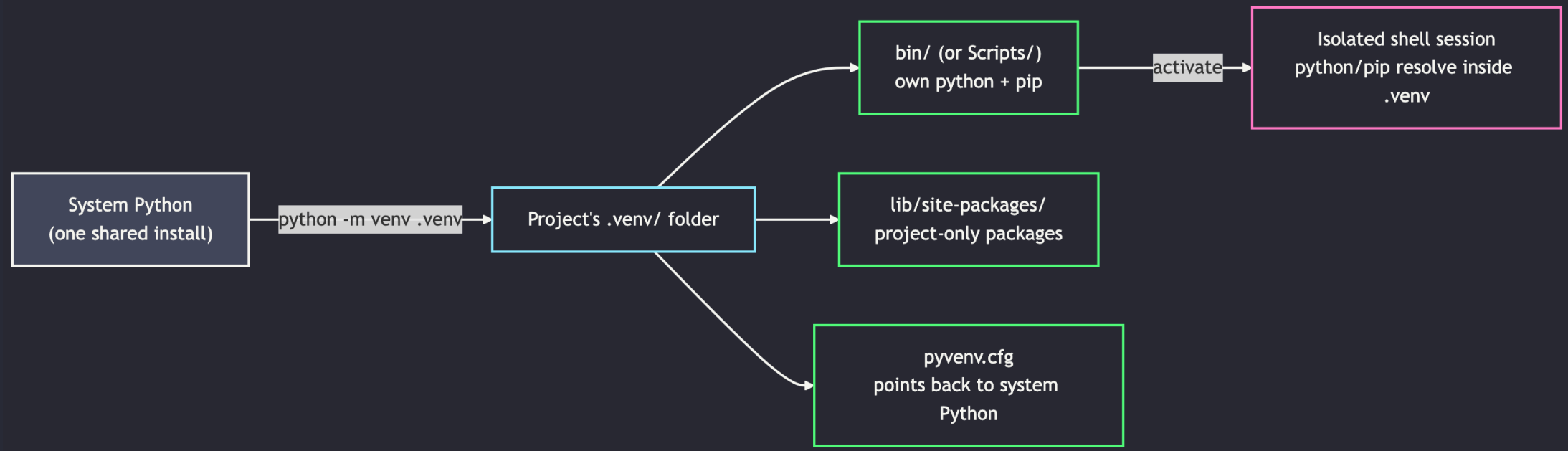
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What a Virtual Environment Actually Is

- A **self-contained folder** holding its own Python interpreter and package library
- Not a separate Python install — it's a lightweight pointer back to your system Python
- Activating it changes *only* what `python` and `pip` resolve to in your current shell
- Deleting the folder removes the environment entirely — nothing else on your machine is affected

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Anatomy of a `venv`



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Tooling Landscape

- `venv` — built into Python 3.3+, zero extra install, what we'll demo today
- `uv` — much faster, manages Python versions too; used across our own course repos
  - Look for `pyproject.toml` + `uv.lock` in a project — that's the `uv` signature
- `conda` — popular in data science, manages non-Python dependencies too (e.g. C libraries)
- `poetry` — adds dependency resolution and packaging on top of venvs
- Today: stick with `venv` — every tool above builds on the same isolation concept

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Demo: Create and Activate

```
# 1. Create a venv in your project folder
python -m venv .venv

# 2. Activate it
source .venv/bin/activate        # macOS / Linux
.venv\Scripts\activate           # Windows (PowerShell/cmd)

# 3. Confirm it's active — prompt shows (.venv), and:
which python                     # macOS / Linux
where python                     # Windows
```

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Demo: Install, Freeze, Deactivate

```
# 4. Install a package — lands inside .venv, not globally
pip install requests

# 5. Snapshot exact versions for reproducibility
pip freeze > requirements.txt

# 6. Leave the venv
deactivate
```

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Demo: Reproduce an Environment

```
# On a new machine, or a fresh clone of the project:
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

- Anyone can recreate the *exact* same environment from `requirements.txt`
- This is what makes a project's setup reproducible for grading, collaboration, or deployment

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Common Pitfalls

- **Forgetting to activate** before `pip install` — package lands in the global environment instead
- **Committing `.venv/` to git** — add it to `.gitignore`; commit `requirements.txt` instead
- **One venv per project** — don't reuse a single venv across unrelated projects
- **A broken venv is disposable** — just `deactivate`, delete the folder, and recreate it

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Cheat Sheet

Action	Command
Create	`python -m venv .venv`
Activate (macOS/Linux)	`source .venv/bin/activate`
Activate (Windows)	`.venv\Scripts\activate`
Install a package	`pip install <package>`
Save exact versions	`pip freeze > requirements.txt`
Reinstall from file	`pip install -r requirements.txt`
Deactivate	`deactivate`
Delete	remove the `.venv/` folder

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References

Topic	Source
`venv` docs	Python Software Foundation. (2026). `venv` — Creation of virtual environments. Python 3 Documentation.
PEP 405	Eby, C. (2012). PEP 405 – Python Virtual Environments. Python Enhancement Proposals.
`pip` guide	Python Packaging Authority. (2026). Installing packages using pip and virtual environments. Python Packaging User Guide.
`uv` docs	Astral. (2026). uv — An extremely fast Python package manager. Astral Docs.

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Recap & Practice

- A venv is a disposable, isolated copy of your project's packages
- Create → activate → install → freeze → deactivate
- **Before you write a single line of code on your next assignment: create a venv first**

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